# NeXusTrade / TopG: Complete Trading Strategy Set & Gate Specification

**Version:** Phase 12.92 Turbo-Refinement

**System Architecture:** Local-First Autonomous Solana Meme-Coin Trading Engine

**Execution Runtime:** TypeScript / Node.js / SQLite (In-Memory)

**Target Performance:** +3.0\% to +6.0\% Daily Net Portfolio Gain (Targeting +5.0\% to +10.0\% Gross)

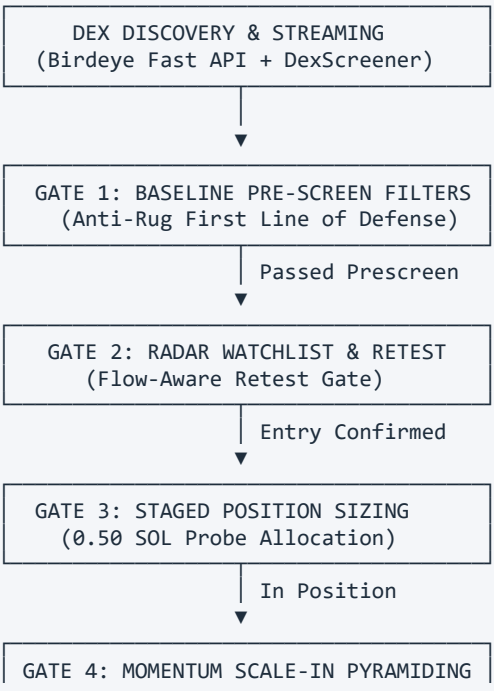
---

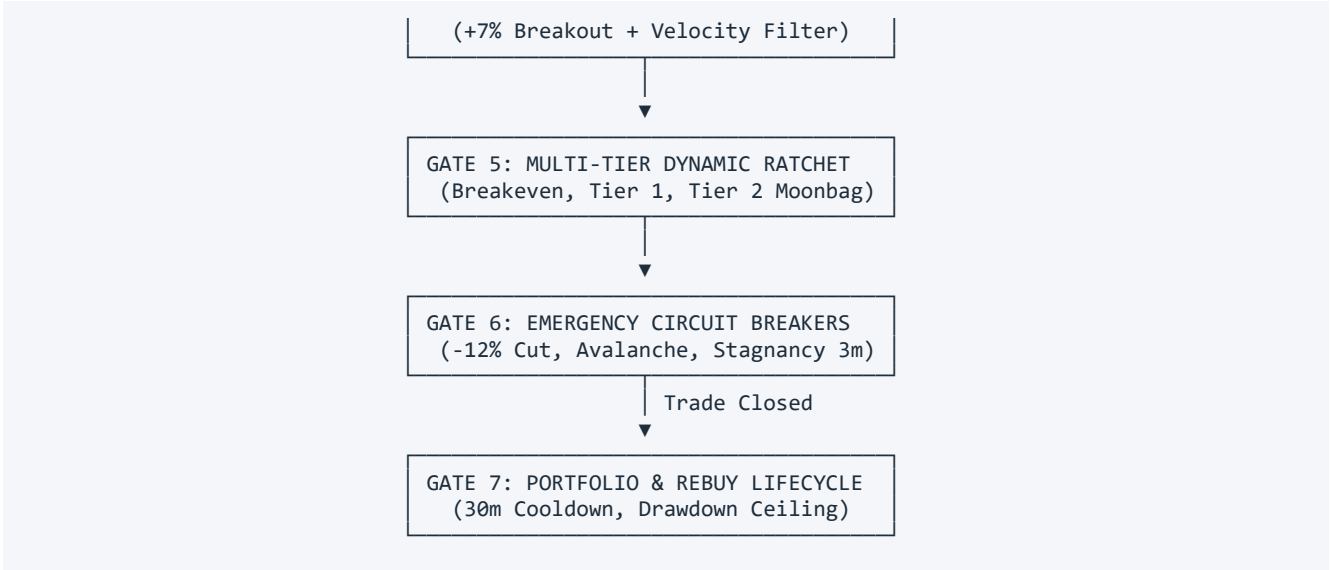
## 1. Executive Strategy Overview

The NeXusTrade autonomous trading architecture operates on a **defense-first, staged-conviction model** tailored specifically for the extreme volatility, predatory liquidity mechanics, and cabal pump-and-dump dynamics of the Solana meme-coin ecosystem (Pump.fun and Raydium CPMM).

The strategy rejects two fatal flaws common in naive crypto trading bots:

1. **Blind Market-Buying**: Never market-buy green candle breakouts at peak euphoria; wait for structural absorption via the Flow-Aware Retest Gate.
2. **Fixed Stop-Loss / Take-Profit Orders**: Fixed -10\% or +20\% orders get front-run or wick-liquidated by MEV bots. NeXusTrade employs an active, stateful multi-tier dynamic ratchet stop-loss engine that moves stops upwards monotonically, takes partial profits in staged tranches, and trails parabolic moonbags with dynamic trailing cushions.






## 2. Gate 1: Baseline Scanner Pre-Screen Filters (Anti-Rug Shield)

Before any token is considered for trade monitoring, it must pass seven non-negotiable structural security and liquidity filters.

### 2.1 Prescreen Gate Parameters

| Filter Parameter | Threshold / Rule | Rationale & Protection |
|---|---|---|
| **Minimum Liquidity (L)** | ≥ \10,000 USD (Ideal: \15k–\50k) | Eliminates micro-pools where a 0.5 SOL buy moves price > 5\% or dev can rug for pennies. |
| **Market Cap Range (MC)** | \40,000 to \500,000 USD | Focuses strictly on sub-million high-velocity meme tokens while discarding pre-curve noise and bloated tops. |
| **Liquidity-to-MC Ratio (L/MC)** | ≥ 15.0\% | Rejects synthetic pump schemes where market cap is spoofed to \2M with only \5k liquidity. |
| **LP Burn / Permanent Lock** | ≥ 90.0\% | Prevents liquidity drain rugs. Unburned LP is immediately rejected. |
| **Mint Authority** | Strictly `null` (Revoked) | Prevents developer from minting infinite tokens and dumping to zero. |
| **Freeze Authority** | Strictly `null` (Revoked) | Prevents honeypots where developer freezes buyer token accounts. |
| **Token Age Window** | 300s to 900s (5 to 15 minutes) | Protects against sniper sandwich attacks in the first 5 minutes while avoiding stale, dying coins older than 15 minutes. |

| | | |
|---|---|---|
| **5-Minute Volume** | ≥ \10,000 USD | Guarantees genuine live retail liquidity turnover. |
| **5-Minute Transactions** | ≥ 25 transactions | Filters low-activity wash-trading pools. |

### 2.2 Disqualification Codes

- `FAILED_BASELINE_SCANNER_PRESCREEN`: Token failed one or more structural filters.
- **Empirical Validation**: Filtered out **103 total rugs** during live sessions, including catastrophic crashes like `NEAR` (-100\%), `USELESS` (-99.96\%), `MBOT` (-99.24\%), and `CLAUD` (-98.83\%).

---

## 3. Gate 2: Radar Watchlist & Flow-Aware Retest Gate

Tokens that pass the baseline prescreen are **never bought at market immediately**. They are placed on the active Radar Watchlist (`armedPullbackCandidates`) to observe structural price action.

### 3.1 Arming Mechanism

When a candidate enters radar:

- `armedAtMs`: Timestamp of arrival.
- `peakPriceSol`: Highest spot price observed while on radar.
- `ticksObserved`: Count of live evaluations.

### 3.2 The Flow-Aware Retest Rules

```
Price from Peak
   0.0%  ──────────────── Peak Observed ────────────────
         │
         │  [Rule B: Consolidation Breakout (0% to -4%)]
         │   • Duration: >= 30s or >= 3 ticks
         │   • Buyer Dominance: >= 1.25x buys over sells
         │   • Status: Confirms Entry!
         │
  -4.0%  ──────────────────────────────────────────────
         │
         │  [Rule A: Retest Pullback Discount (-4% to -10%)]
         │   • Pullback Ratio: 0.90 <= Spot / Peak <= 0.96
         │   • Absorbing Flow: >= 1.15x buys over sells
         │   • Status: Confirms Entry! (CDRIP / UNCC Fix)
         │
 -10.0%  ──────────────────────────────────────────────
         │
         │  [Rule C: Anti-Falling Knife Drop]
         │   • If Flow Absorbing (>= 1.15x): Allowed down to -12.0%
         │   • If Flow Unabsorbed (< 1.15x): DROPPED at -8.0%
         │
 -12.0%  ──────────────────────────────────────────────
         │  > -12.0% Drop: Drop candidate unconditionally (cratered)
```

#### Rule A: Retest Pullback Discount Entry

- **Established Cohort**: Price pulls back between 4.0\% and 10.0\% from peak (`minPullbackRatio = 0.90`, `maxPullbackRatio = 0.96`), with buyer absorption ratio ≥ 1.15x (`recentBuys60s >= 1.15 * recentSells60s`).
- **Micro-Cap Cohort**: Price pulls back between 10.0\% and 15.0\% from peak (`minPullbackRatio = 0.85`, `maxPullbackRatio = 0.90`), with buyer flow ≥ 1.25x.
- **CDRIP / UNCC Case Study**: In Phase 12.91, `CDRIP` had an 8.2\% dip at 4:23 PM with strong buyers (1.5x ratio) before soaring +24,193\%. The old rigid 8.0\% cliff dropped it. Expanding Rule A to 10.0\% with a 12.0\% absorbing ceiling captures these monster runners.

#### Rule B: Consolidation Breakout Entry

- **Established Cohort**: Price holds within 0.0\% to 4.0\% of peak (`maxConsolidationDropPct = 0.04`) for ≥ 30 seconds or ≥ 3 ticks, with buyer dominance ≥ 1.25x.
- **Micro-Cap Cohort**: Price holds within 0.0\% to 5.0\% of peak for ≥ 30 seconds or ≥ 3 ticks, with buyer dominance ≥ 1.50x.

#### Rule C: Crater Ceiling (Falling Knife Filter)

- If buyer flow is absorbing (≥ 1.15x ratio), ceiling is expanded to **12.0\% dip**.
- If buyer flow is unabsorbed (< 1.15x ratio), candidate drops immediately at **8.0\% dip** (`REJECTED_EXCESSIVE_PULLBACK`).

#### Rule E: In-Flight Seller Surge Circuit Breaker

- If sellers dump while candidate is awaiting pullback discount (≥ 25 sells in 60s and sells ≥ 2.5x buys), disarm candidate immediately (`REJECTED_IN_FLIGHT_SELLER_SURGE`).

#### Rule D: Radar Expiry Timeout

- Established tokens expire after 120 seconds without entry confirmation (`REJECTED_PULLBACK_EXPIRED`).
- Micro-cap tokens expire after 90 seconds.

---

## 4. Gate 3: Capital Allocation & Staged Probe Sizing

Risk is strictly partitioned through fixed probe sizing rather than all-in market entries.

### 4.1 Sizing Specification

| Portfolio Parameter | Established Cohort | Micro-Cap Cohort | Description |
|---|---|---|---|
| **Initial Probe Size** | 0.50 SOL | 0.25 SOL | Small toehold probe to verify live execution and initial tick direction. |
| **Dynamic Wallet Scaling** | \min(0.65, 0.50 × scale) | Flat 0.25 SOL | Scales up to 0.65 SOL as portfolio compounds above 10.0 SOL. |
| **Max Concurrent Positions** | 5 positions | 5 positions | Hard ceiling on simultaneous open exposure. |

| | | | |
|---|---|---|---|
| **Gas Reserve** | 0.05 SOL | 0.05 SOL | Untouchable reserve for transaction and priority fees. |

## 5. Gate 4: Momentum-Only Scale-In Pyramiding

To capture outsized runners without exposing capital to low-volume fakeout traps, positions only scale in when viral retail momentum is mathematically verified.

### 5.1 Pyramiding Qualification Rules

```
Position Reaches Milestone:
  • Established: +7.0% Gain (700 bps)
  • Micro-Cap:  +10.0% Gain (1000 bps)
                    │
                    ▼
         Is Cash Available > 0.5 SOL?
                    │ Yes
                    ▼
        VIRAL MOMENTUM VERIFICATION GATE:
          1. Recent Buys (60s)  >= 15
          2. Buyer Dominance     >= 2.0x (Buys >= 2.0 * Sells)
          3. 5-Minute Momentum   >= +4.0% (momentum5mBps >= 400)
                    │
          ┌─────────┴──────────┐
          ▼ Yes                ▼ No
 ┌──────────────────┐  ┌──────────────────────┐
 │ SCALE IN APPROVED:│  │ SCALE IN DENIED:     │
 │ Add +0.50 SOL     │  │ Keep Flat 0.50 SOL Probe │
 │ Recalculate Blended│  │ Protects capital against │
 │ Entry Price Basis │  │ fakeout reversal trap!   │
 └──────────────────┘  └──────────────────────┘
```

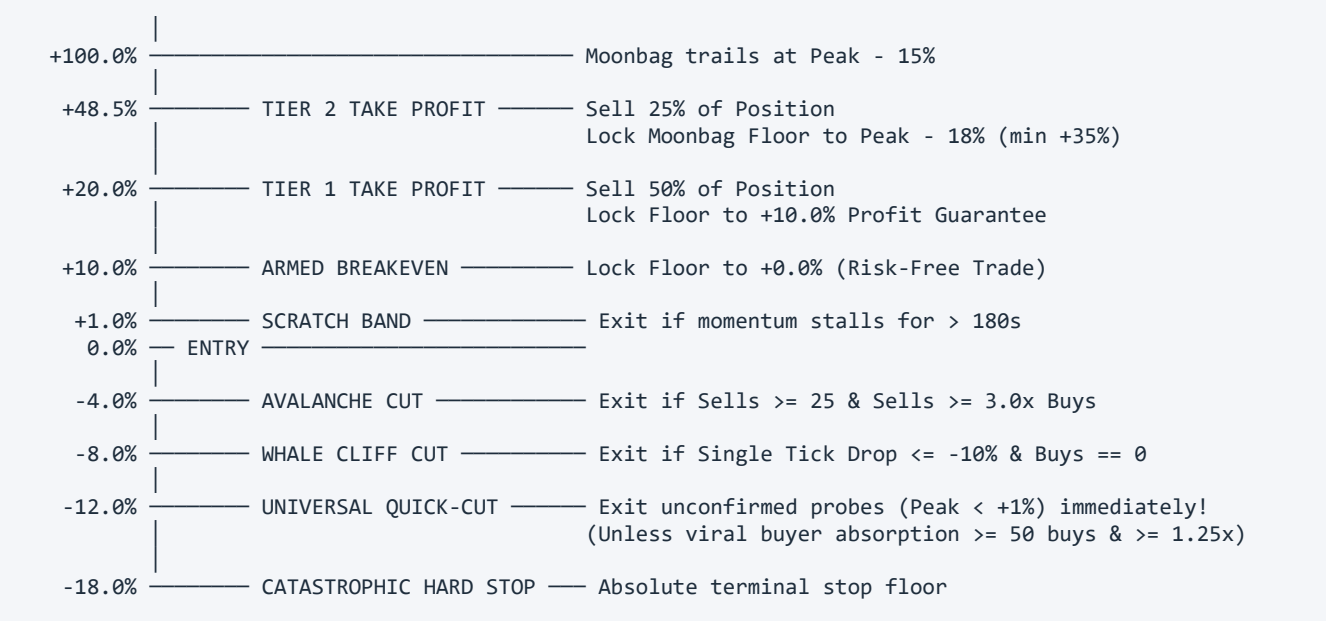
- **Fakeout Reversal Defense**: In Day 2, `CNVQ6V` and `7QB9SH` surged +7\% on low volume (3 buys vs 2 sells) and immediately reversed. The momentum filter denies scale-in, cutting the loss from a doubled 1.0 SOL position down to a flat 0.50 SOL probe.

## 6. Gate 5: Multi-Tier Dynamic Ratchet Stop-Loss & Take-Profit Engine

NeXusTrade uses a continuous stateful ratchet engine (`DynamicRatchetService`) that evaluates position health on every live tick. Stop-loss floors are **strictly monotonic** (they can only stay flat or move up; they can never be lowered).

### 6.1 Ratchet Milestones & Profit Schedule

```
PnL Gain (%)
  +200.0% ──────────────────────────────────── Moonbag trails at Peak - 10%
```




## 6.2 Detailed Trigger Definitions

3. **Armed Breakeven (+10.0% Peak / 1000 bps)**:
- Once a position touches +10.0\% peak gain, `armedBreakeven` triggers. Stop floor moves to `0 bps` (+0.0\%). The trade is guaranteed never to become red. Exit reason: `ARMED_BREAKEVEN_BREACH`.
4. **Tier 1 Take-Profit (+20.0% Peak / 2000 bps)**:
- Action: Partial sale of 50\% of held tokens (`SELL_PARTIAL_50`).
- Stop Floor: Moves to `+10.0%` (1000 bps) on remaining 50\% tokens.
- Exit reason: `RATCHET_TIER_1_BREACH`.
5. **Tier 2 Take-Profit (+48.5% Peak / 4850 bps)**:
- Action: Partial sale of 25\% of original position (`SELL_PARTIAL_25`).
- Stop Floor: Locks to minimum `+35.0%` (3500 bps) or Dynamic Trailing Moonbag.
6. **Tightened Dynamic Trailing Moonbag (Remaining 25% Tokens)**:
- **Base Buffer**: Trails **18.0\% (1800 bps)** below peak (Phase 12.92 tightened from 25%).
- **Parabolic Buffer (≥ +100\% Peak)**: Trails **15.0\% (1500 bps)** below peak.
- **Super-Parabolic Buffer (≥ +200\% Peak)**: Trails **10.0\% (1000 bps)** below peak.
- Exit reason: `RATCHET_TIER_2_BREACH`.
7. **Universal -12% Unconfirmed Quick-Cut (Phase 12.92 Deliverable 2)**:
- Applies to all cohorts (Established and Micro-Cap).
- If position has never demonstrated positive momentum (`peakGainBps < 100`, i.e. < +1.0\%) and drops to ≤ -12.0\% (-1200 bps), it is liquidated immediately via `PROBE_UNCONFIRMED_DUMP_CUT`.
- **Exception**: High-velocity buyer absorption (≥ 50 buys in 60s and buys ≥ 1.25x sells).
- **Empirical Impact**: Stops micro-cap bleeds like `FDEBYP` (which previously fell to -25.9\%), saving 0.14 SOL per trapped position.
8. **Avalanche Sell-Pressure Emergency Cut (Sub-Phase 12.86)**:
- If position is underwater (P/L ≤ -4.0\%) and live seller avalanche hits (≥ 25 sells/min AND sells ≥ 3.0x buys), exit immediately via `EMERGENCY_SELL_PRESSURE_CUT`.
9. **Single-Tick Whale Dump Circuit Breaker**:

- If position is underwater (P/L ≤ -8.0\%), zero buys in 60s, and single-tick cliff drop ≤ -10.0\%, exit immediately via `WHALE_DEV_DUMP_CLIFF_CUT`.

10. **Catastrophic Hard Stop**:

- Established: -18.0\% (-1800 bps).
- Micro-Cap: -20.0\% (-2000 bps).
- Flow-Aware Band (-12\% to catastrophic floor): If buys < sells or buys = 0, cut immediately via `SELL_PRESSURE_UNABSORBED`.

---

# 7. Gate 6: Capital Turnover & 3-Minute Stagnancy Quick-Release

Solana meme-coin momentum is perishable. Capital locked in stalled tokens degrades portfolio turnover and misses emerging runners.

## 7.1 Stagnancy Engine Specification

- **Monitoring**: Tracks tick-level price changes, transaction count, and volume stalling.
- **Inactivity Timeout**: **180 seconds (3 minutes)** (Phase 12.92 reduced from 5 minutes).
- **Trigger Condition**: Position experiences no price change and zero transaction activity for ≥ 180 seconds.
- **Action**: Immediate market liquidation via `STAGNANCY_TIMEOUT_EXIT`. Capital is returned to cash to reload on live scanner candidates.

---

# 8. Gate 7: Portfolio-Level Risk & Lifecycle Controls

## 8.1 Macro Controls

11. **Portfolio Drawdown Circuit Breaker**:

- Hard circuit breaker at -5.0\% (-500 bps) portfolio drawdown halts all new entries.
- May be waived with `--waive-drawdown` during research/empirical evaluation sessions.

12. **Anti-Rebuy Cooldown**:

- 30 minutes (`antiRebuyCooldownMs = 1,800,000ms`) per token mint address. Prevents emotional re-entry, revenge trading, and churn.

13. **Chrono-Regime Gating & Scheduled Sessions**:

- Configurable schedule window via `--schedule HH:MM-HH:MM[:TZ]` (e.g. `07:00-18:00:PDT`).
- Outside trading window: daemon idles in warm-standby (zero buys).
- Window onset (07:00 PDT): buy gates and radar scanning automatically activate.
- Window conclusion (18:00 PDT): open positions automatically cash out cleanly via `SCHEDULE_WINDOW_CONCLUDED`, final analytics archive, and wallet persists.

14. **Continuous Virtual Wallet State**:

- Persistent ledger saved in `.tmp/virtual-wallet.json`. Carries balance seamlessly across sessions, reboots, and days without resetting bankroll.

---

## 9. Comprehensive Strategy Gate Reference Matrix

| Gate # | Strategy Subsystem | Primary Trigger / Condition | Action / Output | Protection / Purpose |
|---|---|---|---|---|
| **1.1** | **Prescreen Liquidity** | L < \10,000 USD | Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN` | Prevents high slippage & micro-rugs |
| **1.2** | **Prescreen Market Cap** | MC < \40k or MC > \500k | Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN` | Focuses on optimal meme growth band |
| **1.3** | **Prescreen L/MC Ratio** | L / MC < 15.0\% | Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN` | Blocks synthetic/inflated valuation schemes |
| **1.4** | **Prescreen LP Burn** | LP Burn < 90.0\% | Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN` | Eliminates liquidity removal rugs |
| **1.5** | **Prescreen Authority** | Mint or Freeze ≠ null | Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN` | Eliminates honeypots & mint-dilution rugs |
| **1.6** | **Prescreen Age Window** | Age < 300s or > 900s | Rejection: `FAILED_BASELINE_SCANNER_PRESCREEN` | Avoids sniper sandwiches & stale tokens |
| **2.1** | **Retest Rule A (Pullback)** | 4\%–10\% dip, buys ≥ 1.15x | Confirms Entry: `CONFIRMED_RETEST_PULLBACK` | Enters structural dips (CDRIP / UNCC) |
| **2.2** | **Retest Rule B (Consolidation)** | 0\%–4\% dip for ≥ 30s, buys ≥ 1.25x | Confirms Entry: `CONFIRMED_CONSOLIDATION_BREAKOUT` | Enters sustained momentum breakout |
| **2.3** | **Retest Rule C (Crater Ceiling)** | Dip > 12\% (or > 8\% unabsorbed) | Drop Candidate: `REJECTED_EXCESSIVE_PULLBACK` | Anti-falling knife protection |
| **2.4** | **Retest Rule E (Seller Surge)** | Sells ≥ 25, sells ≥ 2.5x buys | Disarm Candidate: `REJECTED_IN_FLIGHT_SELLER_SURGE` | Blocks in-flight dump waves |
| **2.5** | **Retest Rule D (Expiry)** | Timeout > 120s (90s micro) | Drop Candidate: `REJECTED_PULLBACK_EXPIRED` | Recycles watchlist radar capacity |
| **3.1** | **Probe Sizing** | New Entry Confirmed | Allocate 0.50 SOL (0.25 SOL micro) | Risk-partitioned probe toehold |
| **4.1** | **Momentum Pyramiding** | Gain ≥ +7\%, buys ≥ 15, ratio ≥ 2.0x | Scale In +0.50 SOL | Maximizes runner conviction size |
| **4.2** | **Fakeout Scale-In Denied** | Gain ≥ +7\%, volume stalls | Deny Scale-In (Hold 0.50 SOL) | Protects against reversal trap |
| **5.1** | **Unconfirmed Quick-Cut** | Peak < +1.0\%, P/L ≤ -12.0\% | Exit All: `PROBE_UNCONFIRMED_DUMP_CUT` | Halts unconfirmed probe bleedouts |

| | | | | |
|---|---|---|---|---|
| **5.2** | **Avalanche Emergency Cut** | P/L ≤ -4.0\%, sells ≥ 25, ratio ≥ 3.0x | Exit All: `EMERGENCY_SELL_PRESSURE_CUT` | Evacuates before total collapse |
| **5.3** | **Whale Cliff Cut** | P/L ≤ -8.0\%, tick ≤ -10\%, buys = 0 | Exit All: `WHALE_DEV_DUMP_CLIFF_CUT` | Instant exit on developer dump |
| **5.4** | **Catastrophic Hard Stop** | P/L ≤ -18.0\% (-20.0\% micro) | Exit All: `CATASTROPHIC_HARD_STOP` | Non-negotiable portfolio loss ceiling |
| **5.5** | **Armed Breakeven** | Peak ≥ +10.0\% | Stop Floor locks to +0.0\% | Eliminates loss on confirmed winner |
| **5.6** | **Tier 1 Take-Profit** | Peak ≥ +20.0\% | Sell 50\%, lock floor at +10.0\% | Secures +10\% to +20\% banked profit |
| **5.7** | **Tier 2 Take-Profit** | Peak ≥ +48.5\% | Sell 25\%, lock floor at +35.0\% | Locks outsized milestone profit |
| **5.8** | **Trailing Moonbag** | Peak > +48.5\%, trails 18\% (15\%/10\%) | Exit Remainder: `RATCHET_TIER_2_BREACH` | Rides parabolic meme expansions |
| **6.1** | **Stagnancy Quick-Release** | Inactive for ≥ 180 seconds | Exit All: `STAGNANCY_TIMEOUT_EXIT` | Frees capital from dead tokens |
| **7.1** | **Anti-Rebuy Cooldown** | Token closed | Cooldown 30 minutes | Stops churn & revenge trading |
| **7.2** | **End-of-Schedule Cashout** | 18:00 PDT reached | Liquidate All: `SCHEDULE_WINDOW_CONCLUDED` | Guarantees overnight cash certainty |

---

*Document compiled and certified for NeXusTrade / TopG production deployment.*